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Paper Review: Jamming and Spoofing Techniques for Drone Neutralization: An Experimental Study

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ABSTRACT: This study explores the use of electronic countermeasures to disrupt communications systems in Unmanned Aerial Vehicles (UAVs), focusing on the protection of airspaces and critical infrastructures such as airports and power stations. The research aims to develop a low-cost, adaptable jamming device using Software Defined Radio (SDR) technology targeting key UAV communication links including geolocation (GNSS), radio control and video transmission. The paper demonstrates that dynamic spoofing combined with no-fly zone enforcement is particularly effective in forcing drones to land or exhibit erratic behavior.

1. INTRODUCTION

With the rapid proliferation of Unmanned Aerial Vehicles (UAVs) in both civilian and military sectors, the need for robust counter-UAV (C-UAV) systems has become critical. This paper, published by MDPI in the journal *Drones*, explores the development of a low-cost, Software Defined Radio (SDR) based system designed to neutralize drones through electronic warfare. Unlike high-end military systems, this research focuses on accessibility and adaptability targeting the three primary communication links: Global Navigation Satellite System (GNSS), Radio Control (RC) and Video Transmission.

The study emphasizes that while kinetic solutions exist, electronic neutralization offers a non-destructive method to mitigate threats. By leveraging open-source frameworks and affordable hardware, the authors demonstrate how unauthorized drone activity can be managed in sensitive areas like airports or power stations.

2. DESIGN SYSTEM AND METHODOLOGY

The methodology utilizes a modular hardware setup consisting of an Adalm-Pluto SDR, a power amplifier and various high-gain antennas. The software implementation relies on GNU Radio to generate jamming and spoofing signals.

The system targets specific frequencies: 1.5 GHz for GNSS interference and 2.4 GHz for Command and Control (C2) links. Two primary strategies are implemented: 1. Barrage Jamming: Creating a high-noise floor across a wide bandwidth to drown out signals. 2. Protocol-Aware Jamming: Targeting specific modulation schemes like BPSK used in drone telemetry. 3. Spoofing: Utilizing GPS-SDR-SIM to generate fake coordinates, distinguishing between static and dynamic trajectories.

3. MATHEMATICAL FRAMEWORK

The effectiveness of the neutralization system is governed by the Link Budget. The total radiated power, expressed as Effective Isotropic Radiated Power (EIRP) is calculated as:

$$EIRP_{dBm} = P_{out} + G_{amp} + G_{ant} - L_{loss} \tag{1}$$

Furthermore, the system aims to minimize the Signal-to-Interference Ratio (SIR) at the UAV receiver:

$$SIR = \frac{P_{signal}}{P_{jammer} + N} \tag{2}$$

4. FIGURES AND TABLES

4.1. Figures

Experimental validation shows that effective neutralization depends on overcoming the target's signal strength at the receiver.

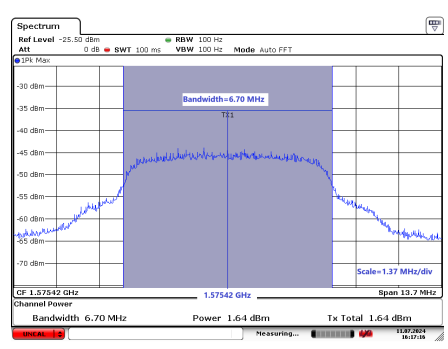

FIGURE 1. GNSS barrage jamming (fc = SR/3) signal spectrum.

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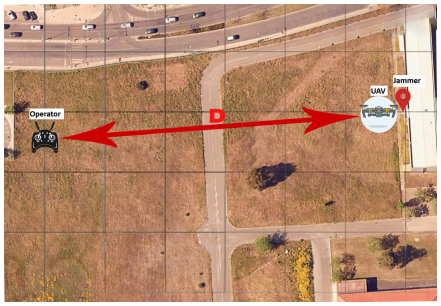

FIGURE 2. Test area at Academia Militar Amadora (Portugal).

4.2. Tables

The following table summarizes the key journal metrics for this publication.

TABLE 1. Journal Performance Metrics (Drones).

Metric	Value	Year
Impact Factor	4.8	2024
CiteScore	7.4	2024
Quartile Ranking	Q1	2024

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